Teaching demonstration

Understanding Inflation

Why Prices Rise and How Central Banks Respond

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Three questions for today

1. What is inflation and how is it measured?
2. Is inflation good or bad?
3. What do central banks do, and at what cost?

A receipt from the same store, 24 years apart



Item	2002	2026	Change
Rice, 5 kg	¥980	¥4,200	+329%
Milk, 1 L	¥98	¥260	+165%
Eggs, 10	¥78	¥310	+297%
Bread, 1 loaf	¥68	¥210	+209%
Cabbage	¥58	¥230	+297%
Total	**¥1,282**	**¥5,210**	**+306%**

Store: JUSCO Hadano, 12 May 2002 → AEON Hadano, 12 Oct 2026

(same supermarket; JUSCO was later renamed AEON)

Average of the five product increases: $(329 + 165 + 297 + 209 + 297) / 5 = \mathbf{259\%}$

Average inflation per year (compound): $\left(\frac{\text{final}}{\text{initial}}\right)^{1/\text{years}} - 1 = \left(\frac{5210}{1282}\right)^{1/24} - 1 \approx \mathbf{6\%}$

From the receipt to the inflation rate

Do you think 6% is the inflation rate?

Milk rose 165%; rice rose 329%.
Whose basket — and how should items be weighted?

The Consumer Price Index

1. Definition

The cost of a fixed basket of goods and services, relative to a base period.

Basket from household spending surveys

(average purchases in the base year).

2. Weights

Weight = category spending ÷ total spending.

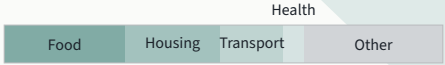
Larger share of spending ⇒ larger weight.

3. Formula

$$\mathrm{CPI}_t = 100 \times \frac{\sum_j p_{tj}\, q_{0j}}{\sum_j p_{0j}\, q_{0j}}$$

j = item · p_{tj} = today's price · q_{0j} = base quantity

What's in the basket



Food	27.5%
Housing	21.8%
Transport	14.4%
Health	4.7%
Other	31.5%

Statistics Bureau of Japan, 2025 base

From the CPI to the inflation rate

CPI = a *level*. Inflation = the *percent change* in that level.

$$\pi_t = \frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}}$$

Example

Year	CPI	π
$t-1$	100	—
t	103	?
$t+1$	105	?

1. Compute both inflation rates.

2. CPI rose again. Is that deflation?

Answers

1. $\pi_t = \dfrac{103-100}{100} = \mathbf{3\%}$

$\pi_{t+1} = \dfrac{105-103}{103} \approx \mathbf{1.9\%}$

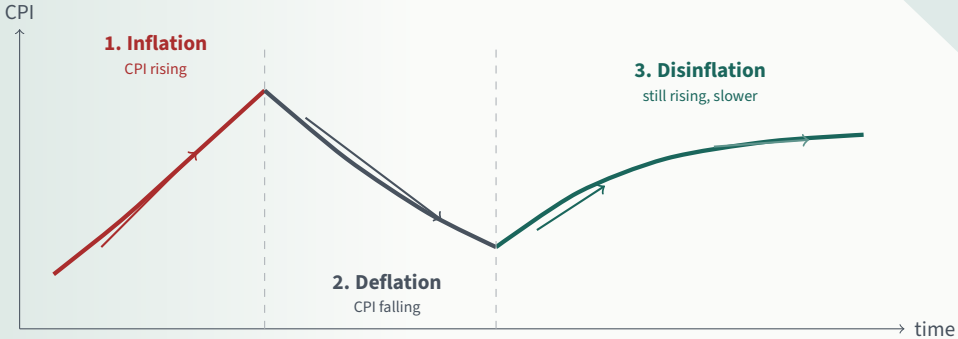
2. No — CPI is still rising.

Inflation fell from 3% to 1.9%: that is **disinflation**.

High / rising level $\neq$ deflation.

Inflation, Deflation, and Disinflation

Vertical axis = CPI (price level). Slope of the curve = inflation. Same idea as 3% → 1.9%.



Disinflation = CPI still up, but the rise is slowing (like 100 → 103 → 105).